

PRIYANSHU DUBEY

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Education

G.L.A University, Mathura – B.Tech, Computer Science (AIML)	2022 – 2026
B.V.P, Agra – Secondary (Class XII), CBSE	2022
B.V.P, Agra – Secondary (Class X), CBSE	2020

Projects

Online Auction Platform

GitHub | Live Demo

React.js, Redux Toolkit, Node.js, Express.js, MongoDB, JWT, Cloudinary

- Developed a multi-role e-commerce auction platform supporting secure authentication and tailored workflows for Buyers, Sellers, and Administrators.
- Architected 15+ highly optimized RESTful APIs using Node.js and Express.js to seamlessly manage low-latency user registration, item creation, and bid transactions.
- Automated the bidding lifecycle by configuring node-cron to instantly settle auctions and freeze bids, eliminating data race conditions during peak high-volume traffic.
- Engineered normalized MongoDB schemas across 5+ data collections and decoupled frontend state using Redux Toolkit to maintain state consistency during real-time UI updates.
- Enforced security protocols by implementing JSON Web Tokens (JWT) and Role-Based Access Control (RBAC) to eliminate unauthorized resource access.

3D Interactive Visualization (React + Zdog)

React.js, JavaScript, Zdog, GSAP, Three.js

- Engineered a lightweight browser-based 3D visualization engine using React and Zdog to programmatically render and scale complex vector-based geometric structures.
- Architected a modular component lifecycle to isolate state mutations, enabling seamless real-time processing of high-frequency user interactions.
- Benchmarked performance at a consistent 60 FPS by synchronizing declarative rendering loops with GSAP timelines, mitigating memory leaks during continuous execution.
- Optimized layout responsiveness for low-resource environments, systematically eliminating GPU rendering bottlenecks across legacy mobile devices.

AI-Powered Fitness Application (Microservices Architecture)

- Engineered a scalable full-stack fitness platform using a microservices architecture to ensure strict modular fault isolation and independent service deployment.
- Designed an API Gateway and integrated dynamic service discovery mechanisms to eliminate single points of failure while optimizing cross-service routing and load balancing.
- Built an event-driven data pipeline using Apache Kafka to handle real-time user activity streaming, successfully reducing backend latency by decoupling heavy processing tasks.
- Implemented robust role-based access control (RBAC) and secure endpoints to manage personalization features, aligning system constraints with production-grade security standards.

Technical Skills

Languages: Java, JavaScript, Python, SQL

Frontend Development: React.js, Redux Toolkit, HTML5, CSS3, GSAP

Backend & Architecture: Node.js, Express.js, Microservices, Apache Kafka, RESTful APIs

Databases & Storage: MongoDB, PostgreSQL, MySQL

Cloud & DevOps: AWS (EC2, S3, IAM, RDS), Docker, Vercel, Render

CS Fundamentals: Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems, Computer Networks

AI & Machine Learning: NumPy, Pandas, Scikit-learn, Supervised Learning, Model Evaluation

Developer Tools: Git, GitHub, Postman, VS Code

Achievements

- Algorithmic Proficiency:** Solved 200+ complex Data Structures & Algorithms (DSA) problems on LeetCode, maintaining a consistent problem-solving cadence.
- System Delivery:** Architected, built, and successfully deployed 3 end-to-end production-ready applications featuring role-based security, state optimization, and real-time streaming architectures.